

Overview of ITU-T G.8031/Y.1342 - Ethernet Linear Protection Switching

Revision history:

Revision	Author	Contact Info	Description	Date
0.1	Yamin		First draft of ELPS learning document	14-08-25
0.2	Yamin		Added 1+1 Unidirectional and Bi-directional protection switching details.	17-08-25
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0.4	Yamin		Added Comparison between protection switching models.	19-08-25
0.5	Yamin		Added APS format	21-08-25

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1. Introduction

Ethernet Linear Protection ensures high availability and fast recovery of Ethernet services when a failure occurs in the network. It protects critical point-to-point Ethernet connections by providing a backup (protection) path in case the primary (working) path fails.

This mechanism is used for VLAN-based Ethernet Sub-Network Connections (SNCs) and is part of the Ethernet Transport (ETH_T) layer in transport networks.

1.1 Key Concepts

- ✓ **Working Transport Entity:** The primary path used to carry normal traffic under normal conditions.
- ✓ **Protection Transport Entity:** The backup path that takes over when the working path fails.
- ✓ **Head End (Bridge Function):** The source node that can duplicate or redirect traffic to either working or protection paths.
- ✓ **Tail End (Selector Function):** The destination node that selects incoming traffic from either working or protection path.
- ✓ **APS (Automatic Protection Switching):** A protocol used between endpoints to coordinate protection switching decisions.
- ✓ **SF (Signal Fail):** A severe defect (e.g., loss of signal) that triggers protection switching. |
- ✓ **SD (Signal Degrade):** A performance degradation (e.g., high error rate) that may trigger switching if configured.

2. Protection Architecture Types

2.1 1+1 Protection

- ✓ *Permanent Bridge:* Traffic is continuously sent on both working and protection paths.
- ✓ *Switching Location:* Only at the receiving end (tail end).
- ✓ *Redundancy:* Full redundancy with dedicated protection resources.
- ✓ *Best For:* Critical services requiring instant failover.

2.2 1:1 Protection

- ✓ *Selective Bridge:* Traffic normally sent only on working path.
- ✓ *Bridge Types Available:*
 - Selector Bridge: Switches traffic to protection path only when needed
 - Broadcast Bridge: Sends traffic on both paths during protection switching
- ✓ *Resource Efficiency:* More efficient as protection bandwidth can be shared.
- ✓ *Best For:* Services requiring good protection with resource optimization.

3. Switching Modes

3.1 Unidirectional Switching:

- Each direction operates independently
- Can protect against failures in opposite directions simultaneously
- No coordination required between endpoints for 1+1

3.2 Bidirectional Switching:

- Both directions switch together, even for single-direction failures
- Requires APS (Automatic Protection Switching) communication for coordination
- Ensures consistent path usage in both directions

4. Revertive vs Non-Revertive Operation

4.1 Revertive Operation:

- Traffic returns to working path after failure is cleared
- Uses WTR (Wait-to-Restore) timer to prevent flapping
- Typical for 1:1 protection

4.2 Non-Revertive Operation:

- Traffic remains on protection path after switching
- Avoids second service disruption when working path is restored
- Common for 1+1 protection

5. Protection Switching Triggers

5.1 Automatic Triggers

1. **Signal Fail (SF)**: Complete loss of signal on working path
2. **Signal Degrade (SD)**: Quality degradation below acceptable threshold

5.2 Manual Triggers

1. **Forced Switch (FS)**: Immediate switch regardless of path condition
2. **Manual Switch (MS)**: Switch when no higher priority conditions exist

6. Role of OAM and APS

6.1 OAM (Operations, Administration, Maintenance):

- ✓ Uses **Continuity Check Messages (CCM)** to monitor health of working and protection paths.

- ✓ MEPs (Maintenance End Points) are activated on both paths for monitoring.

6.2 APS (Automatic Protection Switching):

- ✓ Exchanged over the protection path.
- ✓ Carries commands like "Lockout", "Forced Switch", "Signal Fail", etc.
- ✓ Required for all cases except 1+1 unidirectional switching.

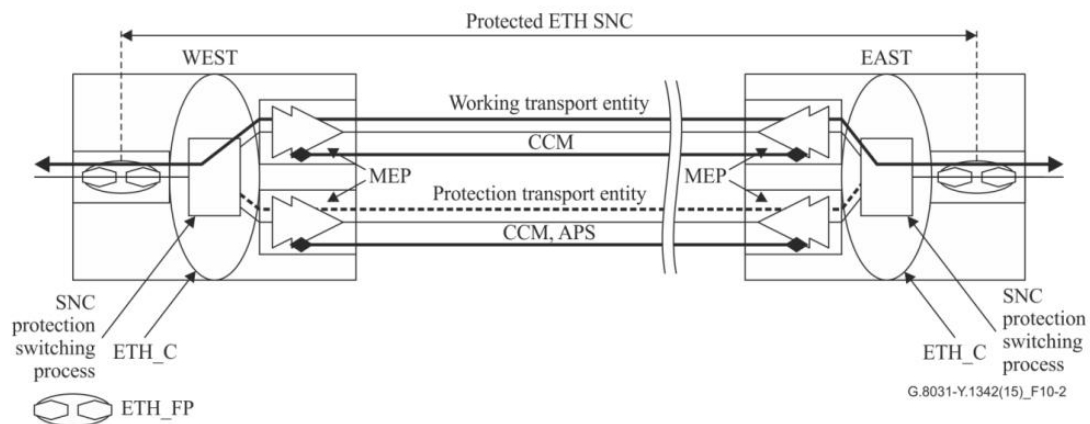
Only 1+1 unidirectional works without APS. It relies solely on local failure detection.

7. Protection switching models

In a protection switching setup, both the working and protection transport entities must be continuously monitored. To achieve this, Maintenance Entity Group End Points (MEPs) are activated on each transport entity.

Each transport entity (working and protection) is monitored by independently exchanging Continuity Check Messages (CCMs) as defined in ITU-T G.8013. This ensures that any defects on either path can be detected promptly.

For protection switching types other than 1+1 unidirectional, an Automatic Protection Switching (APS) communication channel is also required. APS messages are used to coordinate switching decisions between the two ends of the protected domain. These APS Protocol Data Units (PDUs) are transmitted and received over the protection transport entity, using the same MEP pair that exchanges CCMs for link monitoring.



If a MEP detects an anomaly that corresponds to a Signal Fail (SF) condition—or Signal Degrade (SD) if applicable—it reports this failure to the protection switching process.

The functions of terminating CCMs and processing Locked (LCK) signals (also defined in ITU-T G.8013) are performed by the ETH_FT atomic function.

7.1. 1+1 Unidirectional Protection Switching

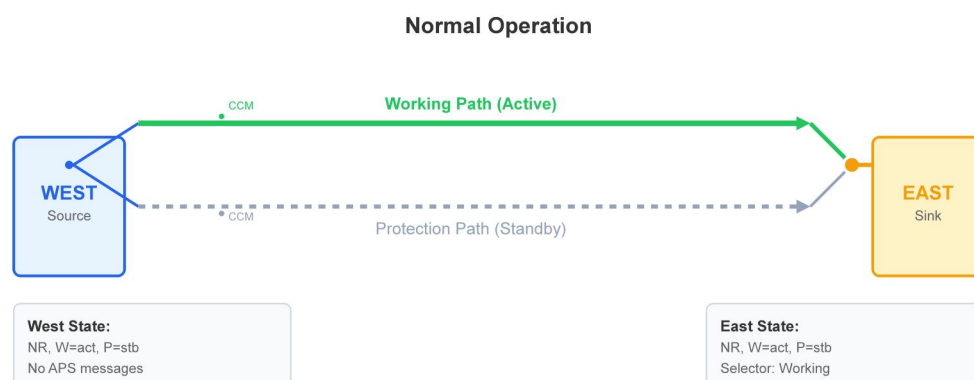
7.1.1 Architecture Overview

- **Bridging:** Permanent — traffic is **always copied** to both working and protection paths at the source.
- **Switching: Unidirectional** — each direction (West-to-East, East-to-West) switches independently.
- **APS Required?** No — switching is based solely on **local defect detection** (e.g., Signal Fail, SF).
- **Selector:** At the receiving end (sink), chooses between working and protection based on signal quality.
- **Typical Mode: Non-revertive** (avoids a second traffic "glitch" when working path recovers).

7.1.2 Initial State (Normal Operation)

- Working path: **Active** (carrying traffic).
- Protection path: **Standby** (receiving traffic, but not selected).
- Selector at East end selects **working path**.
- No APS messages sent (APS not used in unidirectional 1+1).
- CCMs (Continuity Check Messages) are monitored on both working and protection paths.

State: NR (No Request), W=act, P=stb

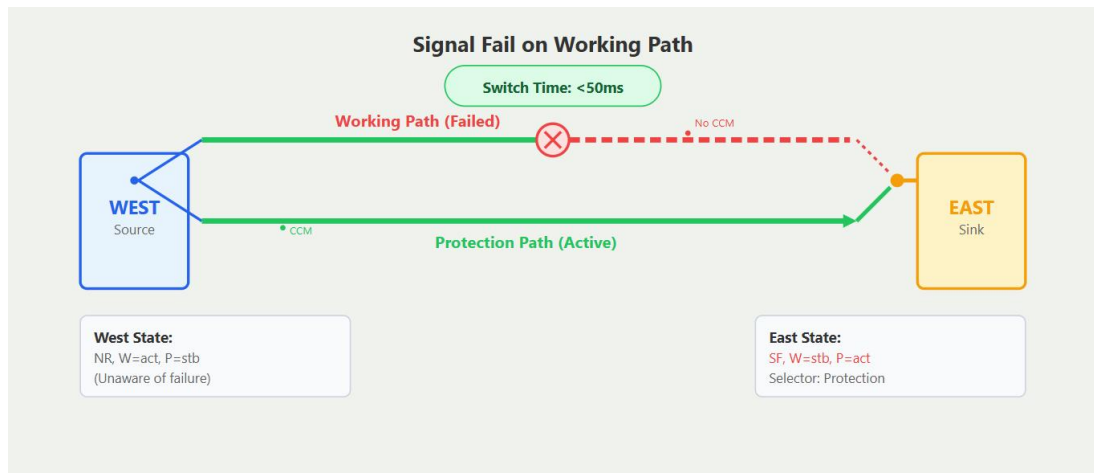


7.1.3 Signal Fail (SF) on Working Path (West-to-East)

- MEP at East detects loss of CCMs on working path → declares **SF(W)**.
- Local priority logic triggers switch to protection.
- Selector at East switches to **protection path**.
- Traffic resumes on protection path (within <50 ms).
- **No APS message is sent** — the West end is unaware of the failure or switch.

State at East: SF, W=stb, P=act

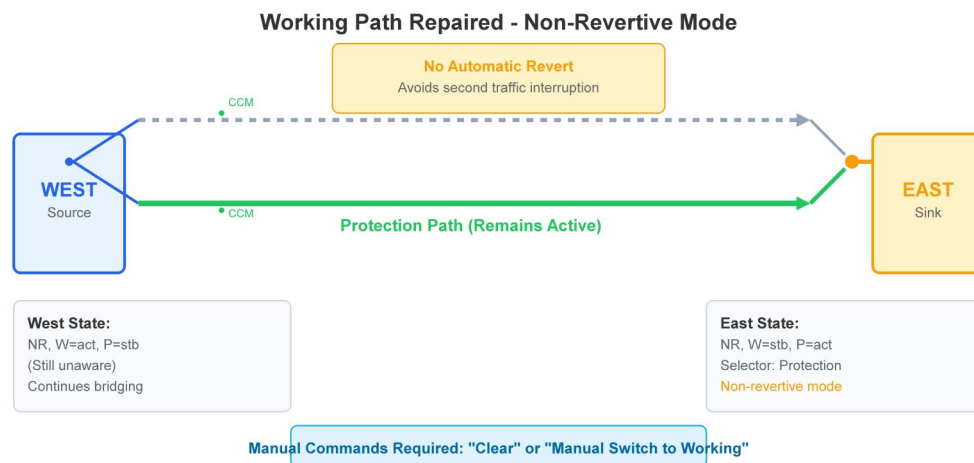
State at West: Still NR, W=act, P=stb (no change)



7.1.4 Working Path Repairs

- SF clears on working path.
- Since this is **non-revertive mode**, the selector **remains on protection**.
- Traffic continues on protection path.
- To return to working, an operator must issue a **Manual Switch to Working** or **Clear** command.

Purpose: Avoids a second traffic interruption ("glitch") during revert.



7.1.5 Key Points

- Fast, autonomous switching — no coordination needed.
- Asymmetric states possible (each end acts independently).
- No APS overhead.
- Ideal for ultra-fast protection where revertive behavior is not required.

7.2. 1+1 Bidirectional Protection Switching

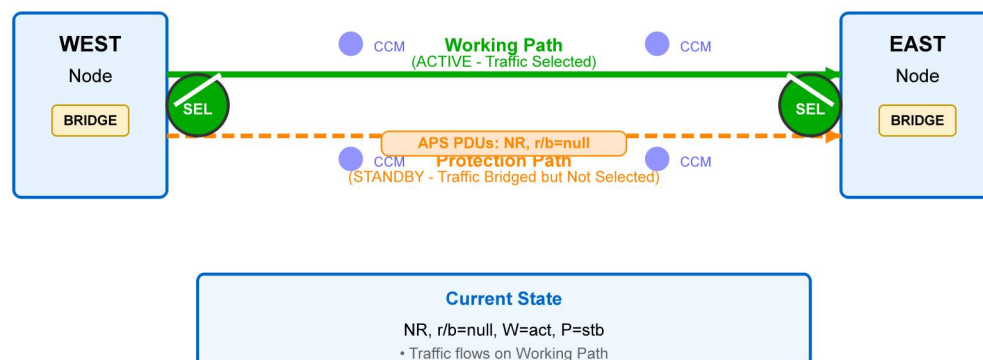
7.2.1 Architecture Overview

- **Bridging:** Permanent — user traffic is **always copied** to both working and protection paths.
- **Switching: Bidirectional** — both directions switch together.
- **APS Required?** Yes — to coordinate selector states at both ends.
- **APS PDU Path:** Sent **only on the protection transport entity**.
- **Selector:** Controlled via APS signaling received on the protection path.
- **Typical Mode:** Can be revertive or non-revertive.

7.2.2 Initial State (Normal Operation)

- Working path: Active (user traffic selected).
- Protection path: Standby (traffic bridged, but not selected).
- Selectors at both ends select **working path**.
- **CCMs** are monitored on **both working and protection paths**.
- **APS PDUs** are exchanged **only over the protection path** (e.g., NR, r/b=null).

State: NR, r/b=null, W=act, P=stb

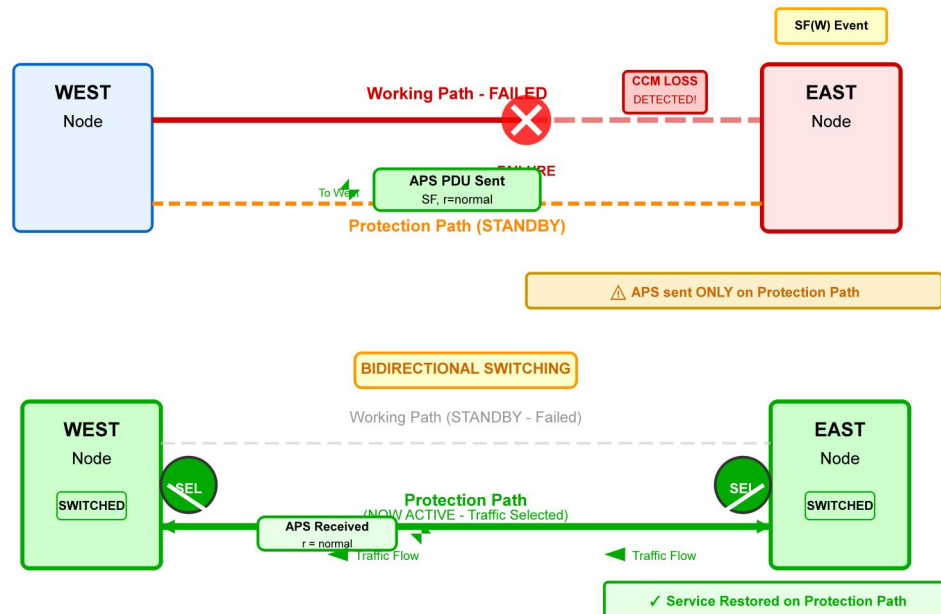


7.2.3 Signal Fail (SF) on Working Path (West-to-East)

1. **East detects SF(W)** via loss of CCMs on working path.
2. **Local priority logic** → highest request = SF.
3. **Global logic** → compares local SF vs. far-end NR (from last APS PDU).
4. SF > NR → decision: **Switch to protection**.
5. **East sends APS PDU:**
 1. **Command:** SF
 2. **Requested Signal (r):** normal (meaning: "select protection")
 3. **Sent over:** Protection transport entity
6. **West receives APS PDU** (via protection path) → sees r = normal.
7. **West sets its selector to protection path.**
8. Both ends now select protection path.

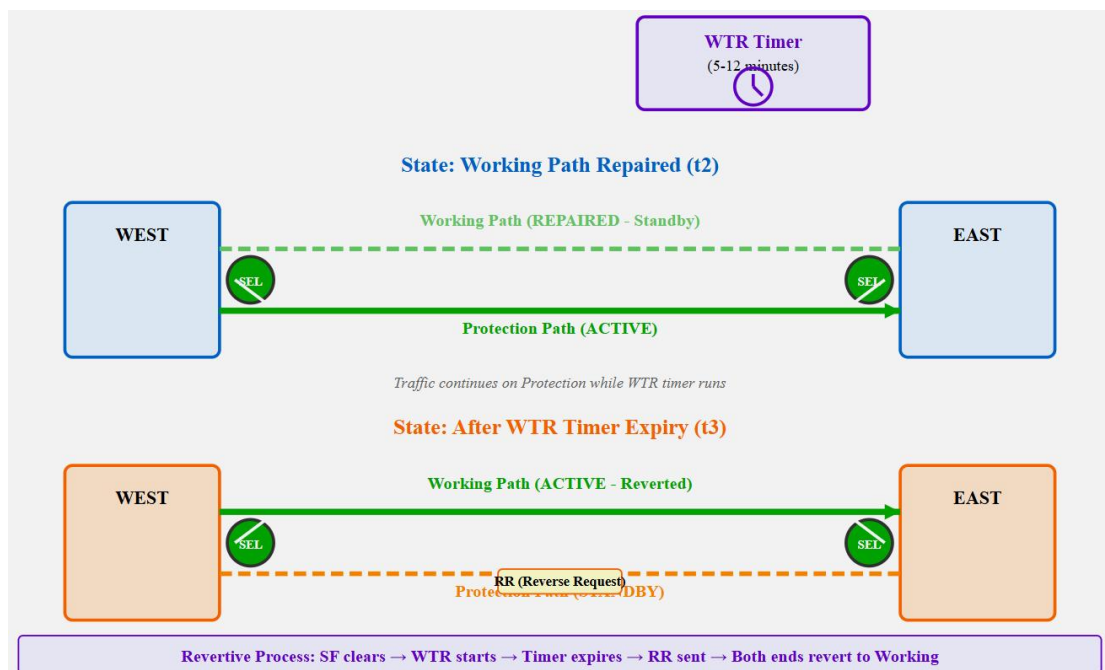
State at both ends: SF, r/b=normal, W=stb, P=act

Important: No APS is sent on the working path. All APS communication occurs on the protection path.

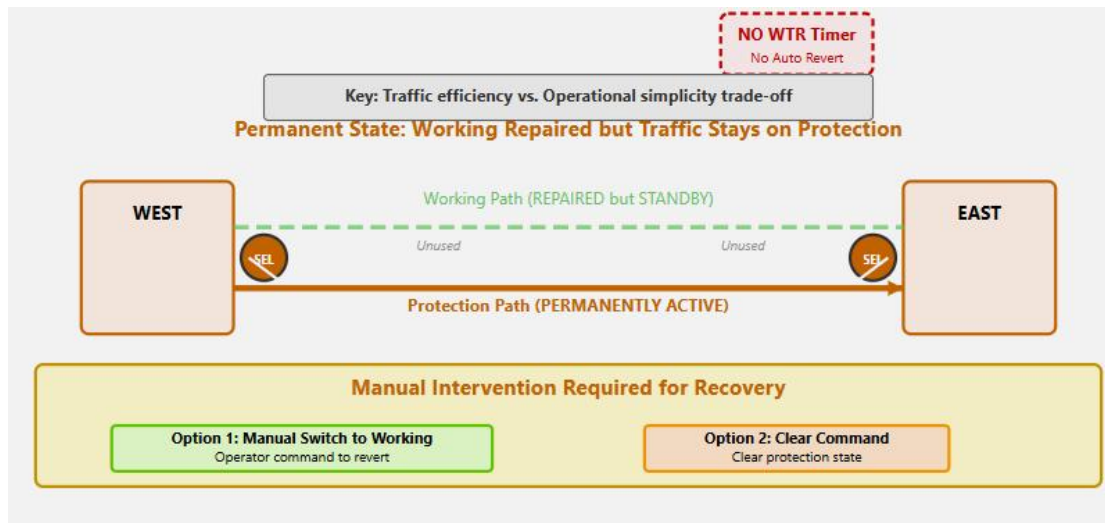


7.2.4 Recovery: Working Path Repairs

- SF clears on working path.
- **If revertive mode:**
 - WTR timer starts (5–12 min).
 - After expiry, RR (Reverse Request) sent via **protection path**.
 - Both ends switch back to working.



- **If non-revertive mode:**
 - Traffic stays on protection.
 - Operator must issue **Manual Switch to Working** or **Clear**.



Key Points

- Both directions switch together — ensures symmetry.
- APS ensures coordination.
- Revertive behavior optional.
- Used when bidirectional consistency is required (e.g., in telecom transport).

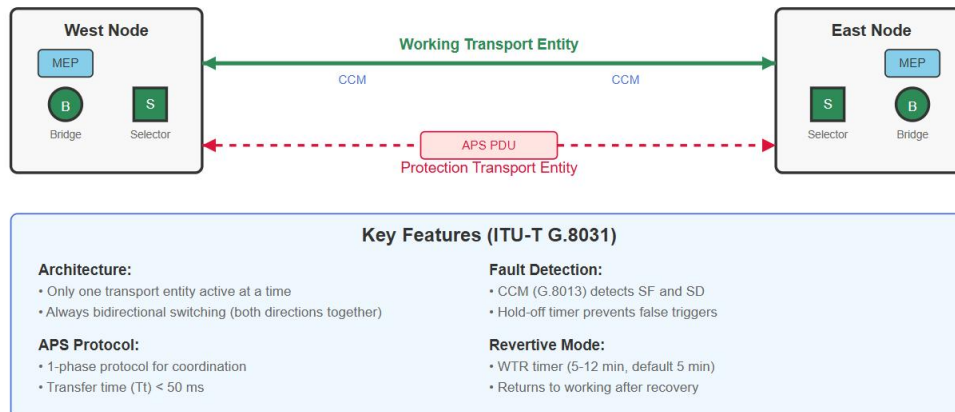
7.3 1:1 Bidirectional Protection Switching

7.3.1 Architecture Overview

- **Bridging:** Not permanent; only one transport entity (working or protection) carries traffic at a time.
- **Switching:** Always bidirectionally; both directions switch together. Requires APS (Automatic Protection Switching) protocol for coordination.
- **Bridge Type:**
 - Selector Bridge: Only the active path carries traffic.
 - Broadcast Bridge: During switching, traffic is temporarily copied to both paths (used for testing or diagnostic purposes).
- **Typical Mode:** Revertive — after recovery of the working path, traffic automatically returns to it after the Wait-to-Restore (WTR) timer expires.
- **Fault Detection:** Relies on Continuity Check Messages (CCM) per ITU-T G.8013 to detect Signal Fail (SF) and Signal Degrade (SD).
- **APS Required:** Yes — 1-phase APS protocol ensures synchronized switching using OAM PDUs sent over the protection transport entity.

Key Point: While APS coordinates the switch, CCM detects the failure. Both are essential.

1:1 Bidirectional Protection Switching (ITU-T G.8031)



7.3.2 Example 1: Signal Fail on Working Path (Revertive Mode)

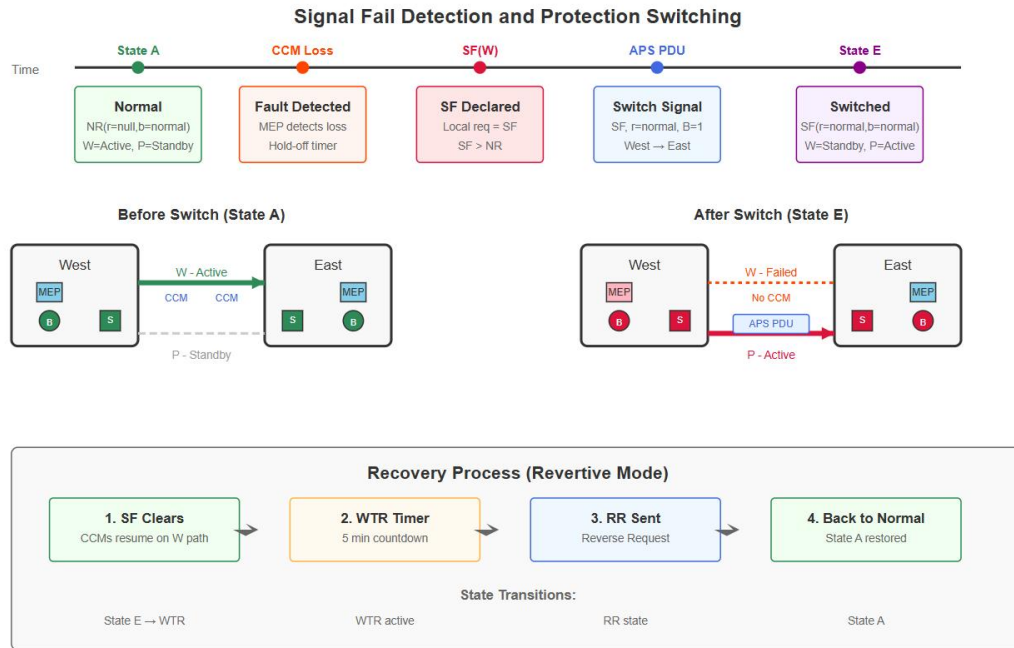
Initial State: A – No Request (NR), Working Active

- State: A
- Signaled APS: NR (r=null, b=normal)
- Working: Active
- Protection: Standby
- CCMs: Normal on both paths
- APS PDUs: Exchanged periodically with NR, r=null, B=1 (revertive)

Signal Fail on Working Path

1. CCM loss on working path detected at West.
2. After hold-off timer, SF(W) declared.
3. Local priority logic triggers switch (SF > NR).
4. APS PDU sent: Command=SF, r=normal, B=1.
5. East receives r=normal → switches selector to protection.
6. West switches bridge to protection.
7. Traffic resumes on protection path.

Result: State → E – SF(r/b=normal).



Recovery: Working Path Repairs

- 1) CCMs on the working path resume at West.
- 2) SF(W) condition is cleared.
- 3) WTR (Wait-to-Restore) timer starts (default 5 minutes, range 5–12 minutes).
- 4) During WTR, protection path continues to carry traffic (flap prevention).
- 5) After WTR expiry, West sends APS PDU: Command=RR, r=null, B=normal.
- 6) East receives RR → switches selector back to working path.
- 7) West switches bridge back to working path.
- 8) Traffic resumes on working path.

Result: State → A – NR(r=null, b=normal).

7.3.3 Example 2: Forced Switch (FS) Command (Revertive Mode)

Initial State: A – NR(r=null, b=normal)

[Working path active, protection path standby.]

1. Operator issues Forced Switch to Protection at West.
2. Local request = FS (highest priority after LO).
3. Global logic compares FS (local) vs. NR (far-end) → FS > NR.
4. APS PDU sent: Command=FS, r=normal, B=1.
5. East receives r=normal → switches selector to protection path.
6. West switches bridge to protection path.

Result: State → D – FS(r/b=normal)

Recovery: Operator Issues "Clear"

1. Operator clears FS at West → local request becomes NR.
2. If revertive mode: WTR timer starts → revert after expiry.
3. If non-revertive mode: stays on protection until operator issues MS to Working.

7.3.4 Example 3: Manual Switch (MS) and Return (Revertive Mode)

Initial State: A – NR($r=\text{null}$, $b=\text{normal}$)

Working path active, protection path standby.

Manual Switch to Protection

1. Operator issues MS at West.
2. Local request = MS.
3. APS PDU sent: Command=MS, $r=\text{normal}$, $B=1$.
4. East receives $r=\text{normal}$ → switches selector to protection path.
5. West switches bridge to protection path.

Result: State → G – MS($r/b=\text{normal}$)

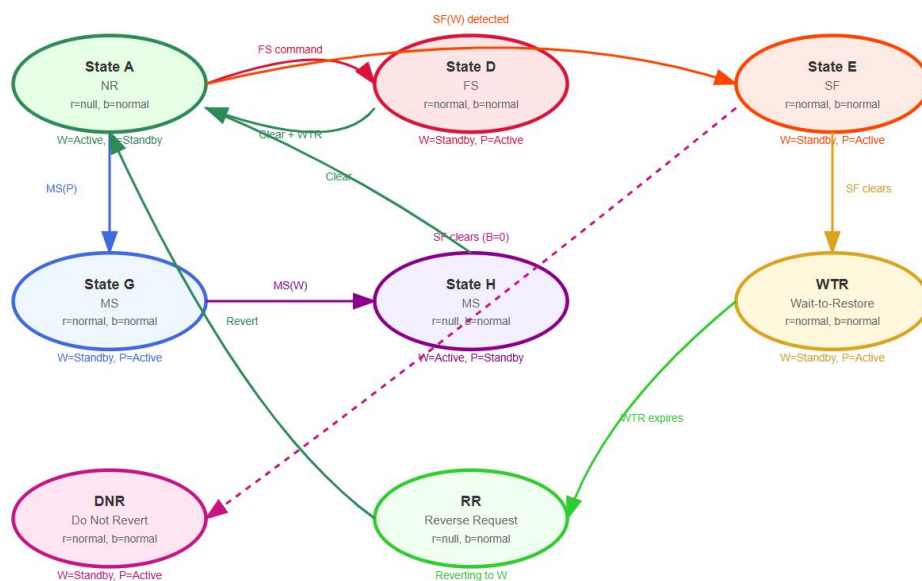
Return to Working

1. Operator issues MS to Working.
2. Local request updates to MS (working).
3. APS PDU sent: Command=MS, $r=\text{null}$, $B=\text{normal}$.
4. East receives $r=\text{null}$ → switches selector to working path.
5. West switches bridge to working path.

Result: State → H – MS($r/b=\text{null}$)

After operator issues Clear, state returns to: A – NR($r=\text{null}$, $b=\text{normal}$).

1:1 Protection State Machine (ITU-T G.8031)



Non-Revertive Mode: Do Not Revert (DNR)

Initial State: A – NR(r=null, b=normal)

1. SF occurs on working → traffic switches to protection.
2. APS PDU: Command=SF, r=normal, B=0 (non-revertive).
3. Traffic resumes on protection path.
4. When working path repairs, no WTR timer starts.
5. Traffic remains on protection.
6. To return to working, operator must issue either:
 - Manual Switch to Working, or
 - Clear (if no higher-priority request is active).

8. Comparison between Protection switching models

Feature	1+1 Unidirectional	1+1 Bidirectional	1:1 Bidirectional
Bridging	Permanent — traffic bridged to both working and protection paths at the source.	Permanent — same as unidirectional.	Selective — only one path (working or protection) carries traffic at a time.
Selector Location	At sink (tail end) only.	At sink (tail end) only.	At sink (tail end).
Bridge Location	Permanent bridge at head.	Permanent bridge at head.	Dynamic bridge at head (switches active path).
Switching Mode	Unidirectional — each direction switches independently.	Bidirectional — both directions switch together, even for unidirectional defects.	Bidirectional — both directions switch together.
APS Required?	No — switching is based solely on local defect detection.	Yes — required for coordinating selectors at both ends.	Yes — required for coordinating bridge and selector actions.
APS PDU Path	Not used (or optional but ignored).	Sent only on protection transport entity. Not sent on working path.	Sent only on protection transport entity. Not sent on working path.
User Traffic on Both Paths?	Yes — always duplicated.	Yes — always duplicated.	No — only on the active path.
CCM on Both Paths?	Yes — for SF/SD monitoring on both working and protection.	Yes — same as unidirectional.	Yes — mandatory to validate path health.
Selector Control	Based on local priority logic only. No far-end input.	Based on global logic: compares local request with far-end APS request (r field).	Based on global logic: both ends coordinate via APS.
State Machine Type	Single-phase, local-only decision. Fast and simple.	Single-phase, coordinated via APS. Ensures symmetry.	Single-phase, coordinated via APS. Full bridge/selector sync.
Revertive Mode	Optional (R=1) — uses WTR timer. Default: often non-revertive.	Optional (R=1) — WTR used if revertive.	Common (R=1) — WTR timer starts after recovery.
Non-Revertive Mode	Common — traffic stays on protection after failure.	Supported — uses DNR (Do Not Revert).	Supported — uses DNR.
Wait-to-Restore (WTR)	Used in revertive mode (5–12 min).	Used in revertive mode.	Used in revertive mode.
Do Not Revert (DNR)	Supported — prevents	Supported — state	Supported — state

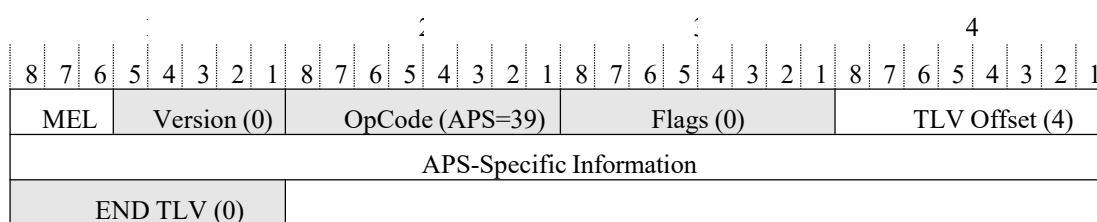
	revert.	DNR(r/b=normal).	DNR(r/b=normal).
Failure on Protection Path (SF-P)	SF-P > FS — prevents switch to protection. Only LO has higher priority.	Same as 1+1 unidirectional.	Same — SF-P blocks switch to protection.
Dual Unidirectional Failures	Can handle (e.g., SF on working west→east, SF on protection east→west).	Cannot handle — both directions forced to same path.	Cannot handle — bidirectional coordination forces symmetry.
Typical Use Case	High availability, fast recovery, asymmetric failure handling.	Symmetric switching, telecom transport where bidirectional consistency is needed.	Bandwidth-efficient protection; protection path can carry extra traffic when idle.
Bandwidth Efficiency	Low — protection path always occupied by duplicate traffic.	Low — same as unidirectional.	High — protection path can carry pre-emptible extra traffic.
Standard Reference	Clause 10.6.2, Table A.9/A.10, Figures IV.33–IV.41	Clause 10.6.1, Table A.5/A.7, Figures IV.17–IV.32	Clause 10.6.3, Table A.1/A.3, Figures IV.1–IV.16
Key Clause	<i>“Unidirectional switching does not need APS coordination protocol.”</i>	<i>“Bidirectional switching needs APS coordination so that selectors for both directions select the same entity.”</i>	<i>“1:1 SNC protection switching can be supported either by a selector bridge or broadcast bridge.”</i>

9. APS format

APS information is carried within the APS PDU which is one of a suite of Ethernet operation, administration and maintenance (OAM) messages. OAM PDU formats for each type of Ethernet OAM operation are defined in [ITU-T G.8013]. APS-specific information is transmitted by a MEP within specific fields in the APS PDU.

The APS messages use the multicast class 1 DA.

The APS PDU is identified by a specific Ethernet OAM OpCode. In this version of the Recommendation, four octets in the APS PDU are used to carry APS-specific information. This is illustrated in Figure 11-1. In addition, it should be noted that for the current version of this Recommendation, the type length value (TLV) Offset field is required to be set to 0x04



9.1 APS PDU format

The Version field shall be set to 0x00.

For other fields such as OpCode, Flags and END TLV, the following values shall be used as defined in [ITU-T G.8013]:

- OpCode:** 0d39 (=0x27)
- Flags:** 0x00
- END TLV:** 0x00

In the MEL field, the MEG level of the APS PDU is inserted.

1								2								3								4								
8	7	6	5	4	3	2	1	8	7	6	5	4	3	2	1	8	7	6	5	4	3	2	1	8	7	6	5	4	3	2	1	
Request/ State				Prot. Type				Requested Signal								Bridged Signal								T	Reserved (0)							
				A	B	D	R																									

9.2 APS-specific information format

All bits defined as "Reserved" in this Recommendation shall be transmitted as 0 and ignored on reception.

Code points and field values

Request/State	1111	Lockout of protection (LO)	Priority
	1110	Signal fail on protection (SF-P)	highest
	1101	Forced switch (FS)	
	1011	Signal fail for working (SF)	
	1001	Signal degrade (SD)	
	0111	Manual switch (MS)	
	0110	Deprecated	
	0101	Wait-to-restore (WTR)	
	0100	Exercise (EXER)	
	0010	Reverse request (RR)	
	0001	Do not revert (DNR)	
	0000	No request (NR)	lowest
	Others	Reserved for future international standardization	

Protection Type	A	0	No APS channel
		1	APS channel
	B	0	1+1 (permanent bridge)
		1	1:1 (no permanent bridge)
	D	0	Unidirectional switching
		1	Bidirectional switching
	R	0	Non-revertive operation
		1	Revertive operation
Requested Signal		0	Null signal
		1	Normal traffic signal
		2-255	(Reserved for future use)
Bridged Signal		0	Null signal
		1	Normal traffic signal
		2-255	(Reserved for future use)
Bridge Type (for 1:1 protection switching)	T	0	Selector bridge
		1	Broadcast bridge

10. Conclusion

ITU-T G.8031/Y.1342 provides a standardized, robust, and efficient framework for Ethernet Linear Protection Switching (ELPS) in point-to-point VLAN-based transport networks. By defining clear mechanisms for 1+1 and 1:1 protection architectures, the recommendation ensures fast, predictable, and operator-controlled service recovery in the event of network failures.

The protocol enables sub-50ms protection switching through a 1-phase APS (Automatic Protection Switching) mechanism, which minimizes signaling overhead while ensuring coordinated state transitions between network endpoints. The integration of OAM functions (CCM) from ITU-T G.8013 allows reliable fault detection on both working and protection transport entities, ensuring that switches occur only to healthy paths.

In summary, ITU-T G.8031/Y.1342 delivers a carrier-grade protection solution tailored for modern Ethernet transport networks. It strikes an optimal balance between performance, reliability, and operational flexibility, making it a foundational standard for building highly available, resilient, and manageable Ethernet infrastructure in metro and core networks.